Java Program

```
import java.util.Scanner;

class HotelBooking
{
   char[][] initializeRooms()
   {
      char[][] rooms = new char[3][5];

      for (int i = 0; i < 3; i++)
      {
         for (int j = 0; j < 5; j++)
         {
            rooms[i][j] = 'A';
         }
      }

      rooms[0][2] = 'B';
      rooms[1][0] = 'B';
      rooms[1][4] = 'B';
      rooms[2][3] = 'B';

      return rooms;
   }

   void displayRooms(char[][] rooms)
   {
      System.out.println("A = Available, B = Booked");

      for (int i = 0; i < 3; i++)
      {
         System.out.print("Floor " + (i + 1) + ": ");

         for (int j = 0; j < 5; j++)
         {
            System.out.print(rooms[i][j] + " ");
         }

         System.out.println();
      }
   }

   void bookRoom(Scanner sc, char[][] rooms)
   {
      int floor;
      int room;
```

```
    System.out.print("Enter floor number (1-3): ");
    floor = sc.nextInt();

    System.out.print("Enter room number (1-5): ");
    room = sc.nextInt();

    if (floor < 1 || floor > 3 || room < 1 || room > 5)
    {
        System.out.println("Invalid floor or room number.");
    }
    else if (rooms[floor - 1][room - 1] == 'B')
    {
        System.out.println("Room is already booked.");
    }
    else
    {
        rooms[floor - 1][room - 1] = 'B';
        System.out.println("Room booked successfully.");
    }
}

public static void main(String[] args)
{
    Scanner sc = new Scanner(System.in);
    HotelBooking hotel = new HotelBooking();
    char[][] rooms = hotel.initializeRooms();
    int choice = 0;

    while (choice != 3)
    {
        System.out.println("\n1. Display Rooms");
        System.out.println("2. Book Room");
        System.out.println("3. Exit");
        System.out.print("Enter your choice: ");

        choice = sc.nextInt();

        switch (choice)
        {
            case 1:
                hotel.displayRooms(rooms);
                break;

            case 2:
                hotel.bookRoom(sc, rooms);
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                break;

            case 3:
                System.out.println("Exiting Hotel Booking System.");
                break;

            default:
                System.out.println("Invalid choice.");
        }
    }

    sc.close();
  }
}
```

Sample Output

1. Display Rooms
2. Book Room
3. Exit

Enter your choice: 1
A = Available, B = Booked
Floor 1: A A B A A
Floor 2: B A A A B
Floor 3: A A A B A

1. Display Rooms
2. Book Room
3. Exit

Enter your choice: 2
Enter floor number (1-3): 3
Enter room number (1-5): 2
Room booked successfully.

1. Display Rooms
2. Book Room
3. Exit

Enter your choice: 2
Enter floor number (1-3): 3
Enter room number (1-5): 2
Room is already booked.

1. Display Rooms
2. Book Room
3. Exit

Enter your choice: 3
Exiting Hotel Booking System.